

An especially interesting feature of the table is that along each diagonal the groups $\pi_{n+k}(S^n)$ with k fixed and varying n eventually become independent of n for large enough n . This stability property is the Freudenthal suspension theorem, proved in §4.2 where we give more extensive tables of these stable homotopy groups of spheres.

Definitions and Basic Constructions

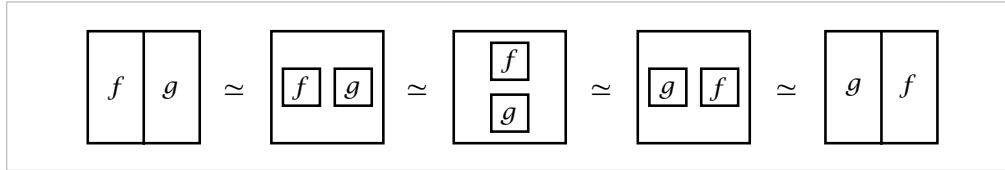
Let I^n be the n -dimensional unit cube, the product of n copies of the interval $[0, 1]$. The boundary ∂I^n of I^n is the subspace consisting of points with at least one coordinate equal to 0 or 1. For a space X with basepoint $x_0 \in X$, define $\pi_n(X, x_0)$ to be the set of homotopy classes of maps $f: (I^n, \partial I^n) \rightarrow (X, x_0)$, where homotopies f_t are required to satisfy $f_t(\partial I^n) = x_0$ for all t . The definition extends to the case $n = 0$ by taking I^0 to be a point and ∂I^0 to be empty, so $\pi_0(X, x_0)$ is just the set of path-components of X .

When $n \geq 2$, a sum operation in $\pi_n(X, x_0)$, generalizing the composition operation in π_1 , is defined by

$$(f + g)(s_1, s_2, \dots, s_n) = \begin{cases} f(2s_1, s_2, \dots, s_n), & s_1 \in [0, 1/2] \\ g(2s_1 - 1, s_2, \dots, s_n), & s_1 \in [1/2, 1] \end{cases}$$

It is evident that this sum is well-defined on homotopy classes. Since only the first coordinate is involved in the sum operation, the same arguments as for π_1 show that $\pi_n(X, x_0)$ is a group, with identity element the constant map sending I^n to x_0 and with inverses given by $-f(s_1, s_2, \dots, s_n) = f(1 - s_1, s_2, \dots, s_n)$.

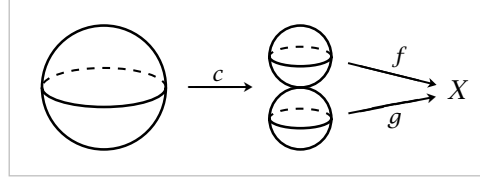
The additive notation for the group operation is used because $\pi_n(X, x_0)$ is abelian for $n \geq 2$. Namely, $f + g \simeq g + f$ via the homotopy indicated in the following figures.



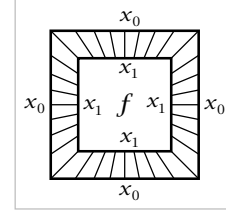
The homotopy begins by shrinking the domains of f and g to smaller subcubes of I^n , with the region outside these subcubes mapping to the basepoint. After this has been done, there is room to slide the two subcubes around anywhere in I^n as long as they stay disjoint, so if $n \geq 2$ they can be slid past each other, interchanging their positions. Then to finish the homotopy, the domains of f and g can be enlarged back to their original size. If one likes, the whole process can be done using just the coordinates s_1 and s_2 , keeping the other coordinates fixed.

Maps $(I^n, \partial I^n) \rightarrow (X, x_0)$ are the same as maps of the quotient $I^n / \partial I^n = S^n$ to X taking the basepoint $s_0 = \partial I^n / \partial I^n$ to x_0 . This means that we can also view $\pi_n(X, x_0)$ as homotopy classes of maps $(S^n, s_0) \rightarrow (X, x_0)$, where homotopies are through maps

of the same form $(S^n, s_0) \rightarrow (X, x_0)$. In this interpretation of $\pi_n(X, x_0)$, the sum $f + g$ is the composition $S^n \xrightarrow{c} S^n \vee S^n \xrightarrow{f \vee g} X$ where c collapses the equator S^{n-1} in S^n to a point and we choose the basepoint s_0 to lie in this S^{n-1} .



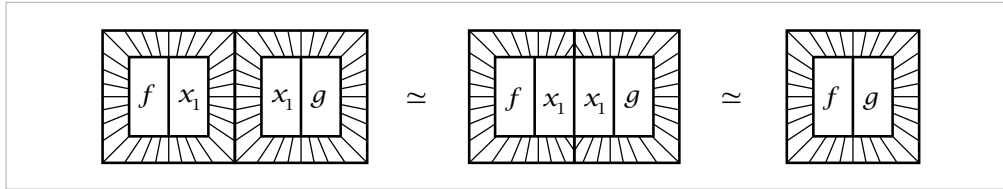
We will show next that if X is path-connected, different choices of the basepoint x_0 always produce isomorphic groups $\pi_n(X, x_0)$, just as for π_1 , so one is justified in writing $\pi_n(X)$ for $\pi_n(X, x_0)$ in these cases. Given a path $\gamma: I \rightarrow X$ from $x_0 = \gamma(0)$ to another basepoint $x_1 = \gamma(1)$, we may associate to each map $f: (I^n, \partial I^n) \rightarrow (X, x_1)$ a new map $\gamma f: (I^n, \partial I^n) \rightarrow (X, x_0)$ by shrinking the domain of f to a smaller concentric cube in I^n , then inserting the path γ on each radial segment in the shell between this smaller cube and ∂I^n . When $n = 1$ the map γf is the composition of the three paths γ , f , and the inverse of γ , so the notation γf conflicts with the notation for composition of paths. Since we are mainly interested in the cases $n > 1$, we leave it to the reader to make the necessary notational adjustments when $n = 1$.



A homotopy of γ or f through maps fixing ∂I or ∂I^n , respectively, yields a homotopy of γf through maps $(I^n, \partial I^n) \rightarrow (X, x_0)$. Here are three other basic properties:

- (1) $\gamma(f + g) \simeq \gamma f + \gamma g$.
- (2) $(\gamma\eta)f \simeq \gamma(\eta f)$.
- (3) $1f \simeq f$, where 1 denotes the constant path.

The homotopies in (2) and (3) are obvious. For (1), we first deform f and g to be constant on the right and left halves of I^n , respectively, producing maps we may call $f + 0$ and $0 + g$, then we excise a progressively wider symmetric middle slab of $\gamma(f + 0) + \gamma(0 + g)$ until it becomes $\gamma(f + g)$:



An explicit formula for this homotopy is

$$h_t(s_1, s_2, \dots, s_n) = \begin{cases} \gamma(f + 0)((2 - t)s_1, s_2, \dots, s_n), & s_1 \in [0, 1/2] \\ \gamma(0 + g)((2 - t)s_1 + t - 1, s_2, \dots, s_n), & s_1 \in [1/2, 1] \end{cases}$$

Thus we have $\gamma(f + g) \simeq \gamma(f + 0) + \gamma(0 + g) \simeq \gamma f + \gamma g$.

If we define a change-of-basepoint transformation $\beta_\gamma: \pi_n(X, x_1) \rightarrow \pi_n(X, x_0)$ by $\beta_\gamma([f]) = [\gamma f]$, then (1) shows that β_γ is a homomorphism, while (2) and (3) imply that β_γ is an isomorphism with inverse $\beta_{\bar{\gamma}}$ where $\bar{\gamma}$ is the inverse path of γ ,

$\bar{y}(s) = y(1-s)$. Thus if X is path-connected, different choices of basepoint x_0 yield isomorphic groups $\pi_n(X, x_0)$, which may then be written simply as $\pi_n(X)$.

Now let us restrict attention to loops y at the basepoint x_0 . Since $\beta_{y\eta} = \beta_y \beta_\eta$, the association $[y] \mapsto \beta_y$ defines a homomorphism from $\pi_1(X, x_0)$ to $\text{Aut}(\pi_n(X, x_0))$, the group of automorphisms of $\pi_n(X, x_0)$. This is called the **action of π_1 on π_n** , each element of π_1 acting as an automorphism $[f] \mapsto [yf]$ of π_n . When $n = 1$ this is the action of π_1 on itself by inner automorphisms. When $n > 1$, the action makes the abelian group $\pi_n(X, x_0)$ into a module over the group ring $\mathbb{Z}[\pi_1(X, x_0)]$. Elements of $\mathbb{Z}[\pi_1]$ are finite sums $\sum_i n_i y_i$ with $n_i \in \mathbb{Z}$ and $y_i \in \pi_1$, multiplication being defined by distributivity and the multiplication in π_1 . The module structure on π_n is given by $(\sum_i n_i y_i)\alpha = \sum_i n_i (y_i \alpha)$ for $\alpha \in \pi_n$. For brevity one sometimes says π_n is a π_1 -module rather than a $\mathbb{Z}[\pi_1]$ -module.

In the literature, a space with trivial π_1 action on π_n is called ‘ n -simple,’ and ‘simple’ means ‘ n -simple for all n .’ It would be nice to have more descriptive terms for these properties. In this book we will call a space **abelian** if it has trivial action of π_1 on all homotopy groups π_n , since when $n = 1$ this is the condition that π_1 be abelian. This terminology is consistent with a long-established usage of the term ‘nilpotent’ to refer to spaces with nilpotent π_1 and nilpotent action of π_1 on all higher homotopy groups; see [Hilton, Mislin, & Roitberg 1975].

We next observe that π_n is a functor. Namely, a map $\varphi: (X, x_0) \rightarrow (Y, y_0)$ induces $\varphi_*: \pi_n(X, x_0) \rightarrow \pi_n(Y, y_0)$ defined by $\varphi_*([f]) = [\varphi f]$. It is immediate from the definitions that φ_* is well-defined and a homomorphism for $n \geq 1$. The functor properties $(\varphi\psi)_* = \varphi_*\psi_*$ and $\mathbb{1}_* = \mathbb{1}$ are also evident, as is the fact that if $\varphi_t: (X, x_0) \rightarrow (Y, y_0)$ is a homotopy then $\varphi_{0*} = \varphi_{1*}$.

In particular, a homotopy equivalence $(X, x_0) \simeq (Y, y_0)$ in the basepointed sense induces isomorphisms on all homotopy groups π_n . This is true even if basepoints are not required to be stationary during homotopies. We showed this for π_1 in Proposition 1.18, and the generalization to higher n ’s is an exercise at the end of this section.

Homotopy groups behave very nicely with respect to covering spaces:

Proposition 4.1. *A covering space projection $p: (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ induces isomorphisms $p_*: \pi_n(\tilde{X}, \tilde{x}_0) \rightarrow \pi_n(X, x_0)$ for all $n \geq 2$.*

Proof: For surjectivity of p_* we apply the lifting criterion in Proposition 1.33, which implies that every map $(S^n, s_0) \rightarrow (X, x_0)$ lifts to $(\tilde{X}, \tilde{x}_0)$ provided that $n \geq 2$ so that S^n is simply-connected. Injectivity of p_* is immediate from the covering homotopy property, just as in Proposition 1.31 which treated the case $n = 1$. $\square$

In particular, $\pi_n(X, x_0) = 0$ for $n \geq 2$ whenever X has a contractible universal cover. This applies for example to S^1 , so we obtain the first row of the table of homotopy groups of spheres shown earlier. More generally, the n -dimensional torus T^n ,

the product of n circles, has universal cover $\mathbb{R}^n$, so $\pi_i(T^n) = 0$ for $i > 1$. This is in marked contrast to the homology groups $H_i(T^n)$ which are nonzero for all $i \leq n$. Spaces with $\pi_n = 0$ for all $n \geq 2$ are sometimes called **aspherical**.

The behavior of homotopy groups with respect to products is very simple:

Proposition 4.2. *For a product $\prod_{\alpha} X_{\alpha}$ of an arbitrary collection of path-connected spaces X_{α} there are isomorphisms $\pi_n(\prod_{\alpha} X_{\alpha}) \approx \prod_{\alpha} \pi_n(X_{\alpha})$ for all n .*

Proof: A map $f: Y \rightarrow \prod_{\alpha} X_{\alpha}$ is the same thing as a collection of maps $f_{\alpha}: Y \rightarrow X_{\alpha}$. Taking Y to be S^n and $S^n \times I$ gives the result. $\square$

Very useful generalizations of the homotopy groups $\pi_n(X, x_0)$ are the **relative homotopy groups** $\pi_n(X, A, x_0)$ for a pair (X, A) with a basepoint $x_0 \in A$. To define these, regard I^{n-1} as the face of I^n with the last coordinate $s_n = 0$ and let J^{n-1} be the closure of $\partial I^n - I^{n-1}$, the union of the remaining faces of I^n . Then $\pi_n(X, A, x_0)$ for $n \geq 1$ is defined to be the set of homotopy classes of maps $(I^n, \partial I^n, J^{n-1}) \rightarrow (X, A, x_0)$, with homotopies through maps of the same form. There does not seem to be a completely satisfactory way of defining $\pi_0(X, A, x_0)$, so we shall leave this undefined (but see the exercises for one possible definition). Note that $\pi_n(X, x_0, x_0) = \pi_n(X, x_0)$, so absolute homotopy groups are a special case of relative homotopy groups.

A sum operation is defined in $\pi_n(X, A, x_0)$ by the same formulas as for $\pi_n(X, x_0)$, except that the coordinate s_n now plays a special role and is no longer available for the sum operation. Thus $\pi_n(X, A, x_0)$ is a group for $n \geq 2$, and this group is abelian for $n \geq 3$. For $n = 1$ we have $I^1 = [0, 1]$, $I^0 = \{0\}$, and $J^0 = \{1\}$, so $\pi_1(X, A, x_0)$ is the set of homotopy classes of paths in X from a varying point in A to the fixed basepoint $x_0 \in A$. In general this is not a group in any natural way.

Just as elements of $\pi_n(X, x_0)$ can be regarded as homotopy classes of maps $(S^n, s_0) \rightarrow (X, x_0)$, there is an alternative definition of $\pi_n(X, A, x_0)$ as the set of homotopy classes of maps $(D^n, S^{n-1}, s_0) \rightarrow (X, A, x_0)$, since collapsing J^{n-1} to a point converts $(I^n, \partial I^n, J^{n-1})$ into (D^n, S^{n-1}, s_0) . From this viewpoint, addition is done via the map $c: D^n \rightarrow D^n \vee D^n$ collapsing $D^{n-1} \subset D^n$ to a point.

A useful and conceptually enlightening reformulation of what it means for an element of $\pi_n(X, A, x_0)$ to be trivial is given by the following *compression criterion*:

- A map $f: (D^n, S^{n-1}, s_0) \rightarrow (X, A, x_0)$ represents zero in $\pi_n(X, A, x_0)$ iff it is homotopic rel S^{n-1} to a map with image contained in A .

For if we have such a homotopy to a map g , then $[f] = [g]$ in $\pi_n(X, A, x_0)$, and $[g] = 0$ via the homotopy obtained by composing g with a deformation retraction of D^n onto s_0 . Conversely, if $[f] = 0$ via a homotopy $F: D^n \times I \rightarrow X$, then by restricting F to a family of n -disks in $D^n \times I$ starting with $D^n \times \{0\}$ and ending with the disk $D^n \times \{1\} \cup S^{n-1} \times I$, all the disks in the family having the same boundary, then we get a homotopy from f to a map into A , stationary on S^{n-1} .

A map $\varphi: (X, A, x_0) \rightarrow (Y, B, y_0)$ induces maps $\varphi_*: \pi_n(X, A, x_0) \rightarrow \pi_n(Y, B, y_0)$ which are homomorphisms for $n \geq 2$ and have properties analogous to those in the absolute case: $(\varphi\psi)_* = \varphi_*\psi_*$, $\mathbb{1}_* = \mathbb{1}$, and $\varphi_* = \psi_*$ if $\varphi \simeq \psi$ through maps $(X, A, x_0) \rightarrow (Y, B, y_0)$.

Probably the most useful feature of the relative groups $\pi_n(X, A, x_0)$ is that they fit into a long exact sequence

$$\cdots \rightarrow \pi_n(A, x_0) \xrightarrow{i_*} \pi_n(X, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial} \pi_{n-1}(A, x_0) \rightarrow \cdots \rightarrow \pi_0(X, x_0)$$

Here i and j are the inclusions $(A, x_0) \hookrightarrow (X, x_0)$ and $(X, x_0, x_0) \hookrightarrow (X, A, x_0)$. The map ∂ comes from restricting maps $(I^n, \partial I^n, J^{n-1}) \rightarrow (X, A, x_0)$ to I^{n-1} , or by restricting maps $(D^n, S^{n-1}, s_0) \rightarrow (X, A, x_0)$ to S^{n-1} . The map ∂ , called the *boundary map*, is a homomorphism when $n > 1$.

Theorem 4.3. *This sequence is exact.*

Near the end of the sequence, where group structures are not defined, exactness still makes sense: The image of one map is the kernel of the next, those elements mapping to the homotopy class of the constant map.

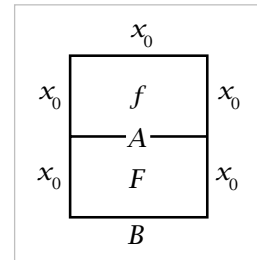
Proof: With only a little more effort we can derive the long exact sequence of a triple (X, A, B, x_0) with $x_0 \in B \subset A \subset X$:

$$\begin{aligned} \cdots \rightarrow \pi_n(A, B, x_0) \xrightarrow{i_*} \pi_n(X, B, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial} \pi_{n-1}(A, B, x_0) \rightarrow \cdots \\ \rightarrow \pi_1(X, A, x_0) \end{aligned}$$

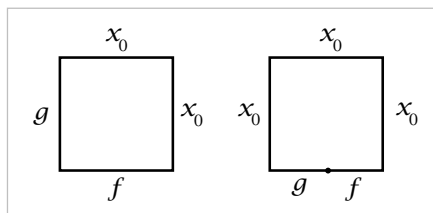
When $B = x_0$ this reduces to the exact sequence for the pair (X, A, x_0) , though the latter sequence continues on two more steps to $\pi_0(X, x_0)$. The verification of exactness at these last two steps is left as a simple exercise.

Exactness at $\pi_n(X, B, x_0)$: First note that the composition j_*i_* is zero since every map $(I^n, \partial I^n, J^{n-1}) \rightarrow (A, B, x_0)$ represents zero in $\pi_n(X, A, x_0)$ by the compression criterion. To see that $\text{Ker } j_* \subset \text{Im } i_*$, let $f: (I^n, \partial I^n, J^{n-1}) \rightarrow (X, B, x_0)$ represent zero in $\pi_n(X, A, x_0)$. Then by the compression criterion again, f is homotopic rel ∂I^n to a map with image in A , hence the class $[f] \in \pi_n(X, B, x_0)$ is in the image of i_* .

Exactness at $\pi_n(X, A, x_0)$: The composition ∂j_* is zero since the restriction of a map $(I^n, \partial I^n, J^{n-1}) \rightarrow (X, B, x_0)$ to I^{n-1} has image lying in B , and hence represents zero in $\pi_{n-1}(A, B, x_0)$. Conversely, suppose the restriction of $f: (I^n, \partial I^n, J^{n-1}) \rightarrow (X, A, x_0)$ to I^{n-1} represents zero in $\pi_{n-1}(A, B, x_0)$. Then $f|_{I^{n-1}}$ is homotopic to a map with image in B via a homotopy $F: I^{n-1} \times I \rightarrow A$ rel ∂I^{n-1} . We can tack F onto f to get a new map $(I^n, \partial I^n, J^{n-1}) \rightarrow (X, B, x_0)$ which, as a map $(I^n, \partial I^n, J^{n-1}) \rightarrow (X, A, x_0)$, is homotopic to f by the homotopy that tacks on increasingly longer initial segments of F . So $[f] \in \text{Im } j_*$.



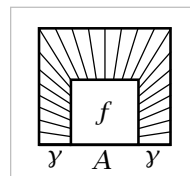
Exactness at $\pi_n(A, B, x_0)$: The composition $i_*\partial$ is zero since the restriction of a map $f: (I^{n+1}, \partial I^{n+1}, J^n) \rightarrow (X, A, x_0)$ to I^n is homotopic rel ∂I^n to a constant map via f itself. The converse is easy if B is a point, since a nullhomotopy $f_t: (I^n, \partial I^n) \rightarrow (X, x_0)$ of $f_0: (I^n, \partial I^n) \rightarrow (A, x_0)$ gives a map $F: (I^{n+1}, \partial I^{n+1}, J^n) \rightarrow (X, A, x_0)$ with $\partial([F]) = [f_0]$. Thus the proof is finished in this case. For a general B , let F be a nullhomotopy of $f: (I^n, \partial I^n, J^{n-1}) \rightarrow (A, B, x_0)$ through maps $(I^n, \partial I^n, J^{n-1}) \rightarrow (X, B, x_0)$, and let g be the restriction of F to $I^{n-1} \times I$, as in the first of the two pictures below. Reparametrizing the n^{th} and $(n+1)^{\text{st}}$ coordinates as shown in the second picture, we see that f with g tacked on is in the image of ∂ . But as we noted in the preceding paragraph, tacking g onto f gives the same element of $\pi_n(A, B, x_0)$. $\square$



Example 4.4. Let CX be the cone on a path-connected space X , the quotient space of $X \times I$ obtained by collapsing $X \times \{0\}$ to a point. We can view X as the subspace $X \times \{1\} \subset CX$. Since CX is contractible, the long exact sequence of homotopy groups for the pair (CX, X) gives isomorphisms $\pi_n(CX, X, x_0) \approx \pi_{n-1}(X, x_0)$ for all $n \geq 1$. Taking $n = 2$, we can thus realize any group G , abelian or not, as a relative π_2 by choosing X to have $\pi_1(X) \approx G$.

The long exact sequence of homotopy groups is clearly natural: A map of basepointed triples $(X, A, B, x_0) \rightarrow (Y, C, D, y_0)$ induces a map between the associated long exact sequences, with commuting squares.

There are change-of-basepoint isomorphisms β_γ for relative homotopy groups analogous to those in the absolute case. One starts with a path γ in $A \subset X$ from x_0 to x_1 , and this induces $\beta_\gamma: \pi_n(X, A, x_1) \rightarrow \pi_n(X, A, x_0)$ by setting $\beta_\gamma([f]) = [\gamma f]$ where γf is defined as in the picture, by placing a copy of f in a smaller cube with its face I^{n-1} centered in the corresponding face of the larger cube. This construction satisfies the same basic properties as in the absolute case, with very similar proofs that we leave to the exercises. Separate proofs must be given in the two cases since the definition of γf in the relative case does not specialize to the definition of γf in the absolute case.



The isomorphisms β_γ show that $\pi_n(X, A, x_0)$ is independent of x_0 when A is path-connected. In this case $\pi_n(X, A, x_0)$ is often written simply as $\pi_n(X, A)$.

Restricting to loops at the basepoint, the association $\gamma \mapsto \beta_\gamma$ defines an action of $\pi_1(A, x_0)$ on $\pi_n(X, A, x_0)$ analogous to the action of $\pi_1(X, x_0)$ on $\pi_n(X, x_0)$ in the absolute case. In fact, it is clear from the definitions that $\pi_1(A, x_0)$ acts on the whole long exact sequence of homotopy groups for (X, A, x_0) , the action commuting with the various maps in the sequence.

A space X with basepoint x_0 is said to be **n -connected** if $\pi_i(X, x_0) = 0$ for $i \leq n$. Thus 0-connected means path-connected and 1-connected means simply-connected. Since n -connected implies 0-connected, the choice of the basepoint x_0 is not significant. The condition of being n -connected can be expressed without mention of a basepoint since it is an easy exercise to check that the following three conditions are equivalent.

- (1) Every map $S^i \rightarrow X$ is homotopic to a constant map.
- (2) Every map $S^i \rightarrow X$ extends to a map $D^{i+1} \rightarrow X$.
- (3) $\pi_i(X, x_0) = 0$ for all $x_0 \in X$.

Thus X is n -connected if any one of these three conditions holds for all $i \leq n$. Similarly, in the relative case it is not hard to see that the following four conditions are equivalent, for $i > 0$:

- (1) Every map $(D^i, \partial D^i) \rightarrow (X, A)$ is homotopic rel ∂D^i to a map $D^i \rightarrow A$.
- (2) Every map $(D^i, \partial D^i) \rightarrow (X, A)$ is homotopic through such maps to a map $D^i \rightarrow A$.
- (3) Every map $(D^i, \partial D^i) \rightarrow (X, A)$ is homotopic through such maps to a constant map $D^i \rightarrow A$.
- (4) $\pi_i(X, A, x_0) = 0$ for all $x_0 \in A$.

When $i = 0$ we did not define the relative π_0 , and (1)–(3) are each equivalent to saying that each path-component of X contains points in A since D^0 is a point and ∂D^0 is empty. The pair (X, A) is called **n -connected** if (1)–(4) hold for all $i \leq n$, $i > 0$, and (1)–(3) hold for $i = 0$.

Note that X is n -connected iff (X, x_0) is n -connected for some x_0 and hence for all x_0 .

Whitehead's Theorem

Since CW complexes are built using attaching maps whose domains are spheres, it is perhaps not too surprising that homotopy groups of CW complexes carry a lot of information. Whitehead's theorem makes this explicit:

Theorem 4.5. *If a map $f: X \rightarrow Y$ between connected CW complexes induces isomorphisms $f_*: \pi_n(X) \rightarrow \pi_n(Y)$ for all n , then f is a homotopy equivalence. In case f is the inclusion of a subcomplex $X \hookrightarrow Y$, the conclusion is stronger: X is a deformation retract of Y .*

The proof will follow rather easily from a more technical result that turns out to be very useful in quite a number of arguments. For convenient reference we call this the **compression lemma**.

Lemma 4.6. *Let (X, A) be a CW pair and let (Y, B) be any pair with $B \neq \emptyset$. For each n such that $X - A$ has cells of dimension n , assume that $\pi_n(Y, B, y_0) = 0$ for all $y_0 \in B$. Then every map $f: (X, A) \rightarrow (Y, B)$ is homotopic rel A to a map $X \rightarrow B$.*